AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2019

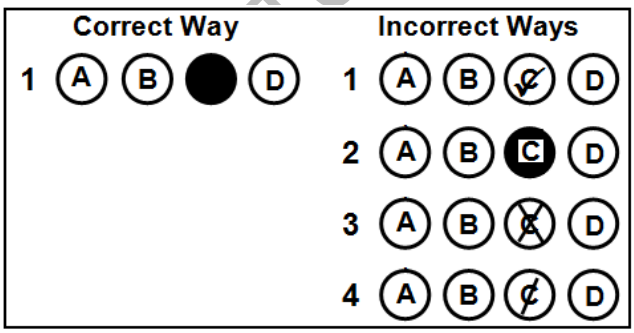
Business Statistics Paper I

Time: 30 minutes Marks: 20

April/ May 2019 for Teaching & Learning only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 20 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. Which of the following statements is NOT the limitation of Statistics?

   A. It is liable to be misused.
   B. It gives results that are only used on average.
   C. It measures the characteristics of population.
   D. It needs conversion of qualitative data into quantitative data.

2. An example of discrete variable is

   A. height of students in a class.
   B. weight of students in a class.
   C. number of students in a class.
   D. temperature of a classroom.

3. A school wants to estimate the average mathematics' score of a group of 500 students for a 75 marks test. For a random sample of 50 students, the average score is 40 marks. The total number of students in the group represents a

   A. sample.
   B. statistic.
   C. parameter.
   D. population.

Use the given information to answer Q.4 and Q.5.

The price and the number of books sold at a book store in the month of June are recorded in the given table.

Price Range (Rs)	**Number of Books Sold**	**Relative Frequency (%)**
150 – 199	66	13.2
200 – 249	?	30
250 – 299	159	31.8
300 – 349	?	?
350 – 399	40	8
Total	**500**	

4. The percentage of books sold, whose price ranges from Rs 300 to 349

   A. is 10%
   B. is 17%
   C. is 83%
   D. cannot be determined.

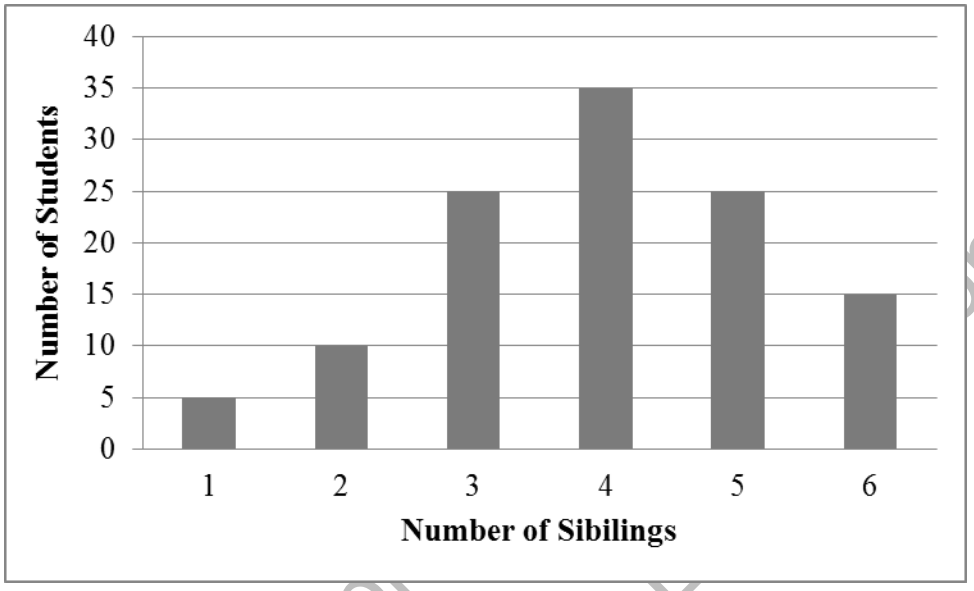
April/ May 2019 for Teaching & Learning only

5. The number of books sold, whose price ranges from Rs 150 to 299

A. is 150
B. is 225
C. is 375
D. cannot be determined.

Use the given information to answer Q.6 and Q.7.

The given bar graph represents the number of students and their siblings in a school.



6. The number of students that have more than three siblings is

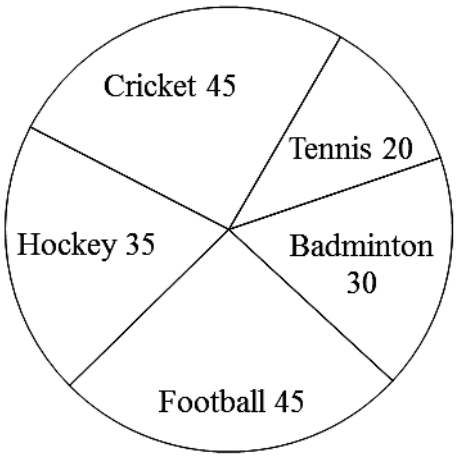
A. 35
B. 40
C. 75
D. 100

7. The number of students that have three or four siblings is

A. 25
B. 35
C. 50
D. 60

Teaching & Learning Only

8. The given pie chart represents information about the sports festival of a school. It shows the number of students who participated in different sports. The angle of the sector representing hockey is



A. 22°
B. 36°
C. 72°
D. 90°

9. The given table shows information about the weight of students at the Higher Secondary School Certificate (HSSC) level.

	HSSC-I	**HSSC-II**
Mean Wight (kg)	53	60
Number of Students	30	25

The total weight of all HSSC students is

A. 6,215 kg.
B. 3,125 kg.
C. 3,090 kg.
D. 113 kg.

10. The mean weight of twenty objects was calculated to be 196 grams. However, it was found later that due to an error in the weighing machine, the weight of each object had been recorded with an increase of 3 grams. The correct mean weight of twenty objects, therefore, will be

A. 136
B. 193
C. 196
D. 199

Apr/ May 2019 Teaching & Learning only

Use the given information to answer Q.11 and Q.12.

The table shows information about the height of 120 buildings in a locality.

Height (ft)	Number of Buildings
26 – 35	9
36 – 45	17
46 – 55	30
56 – 65	40
66 – 75	14
76 – 85	10

11. The frequency of the median class is

A. 96
B. 56
C. 40
D. 30

12. The lower class boundary of the modal class is

A. 55
B. 55.5
C. 56
D. 56.5

13. For the given raw data, 14, 6, 10, 12, 8, 9, 11, the sum of deviations from the mean is

A. 0
B. 10
C. 14
D. 70

14. If the standard deviation of a, b, c, d, e is 4, then the standard deviation of $2a-5, 2b-5, 2c-5, 2d-5, 2e-5$ will be

A. 3
B. 4
C. 8
D. 16

April/ May 2019 for Teaching & Learning only

15. The table gives information about the train fare from Karachi to Lahore. All information is of the same class and of the same train.

Year	Price (Rs)
2010	2,900
2011	3,100
2012	3,250
2013	3,500
2014	3,800

Taking 2010 as the base period, the price index of year 2013 indicates that the train fare is increased by

A. 17.14%
B. 20.69%
C. 82.86%
D. 120.69%

16. The given table provides information about the prices of two commodities and their quantities sold in 2017 and 2018, respectively.

Commodity	2017			2018		
	Price (P_o) (Rs)	Quantity (Q_o) (kg)	P_oQ_o	Price (P_1) (Rs)	Quantity (Q_1) (kg)	P_1Q_0
P	1	3	3	3	2	9
Q	2	3	6	4	3	12
Total	**3**	**6**	**9**	**7**	**5**	**21**

The Laspeyre's weighted index will be

A. 42.86
B. 83.33
C. 120
D. 233.33

17. The number of possible ways to choose 3 students from a class of 18 will be

A. 54
B. 816
C. 5,832
D. 4,896

April May 2019 Teaching & learning only

18. Ali and Sara are interested `in buying a computer. They have the following options:

- 3 types of keyboards
- 5 types of monitors
- 5 types of desktops
- 2 types of mouse

The number of possible choices for them to buy a computer is

A. 15
B. 15!
C. 150
D. 150!

19. A box contains 10 slips of identical size. The slips are numbered from 1 to 10.
If a slip is drawn at random from the box, then the probability of getting a slip containing a number that is even and a multiple of three will be

A. 0
B. $\frac{1}{10}$
C. $\frac{5}{10}$
D. $\frac{6}{10}$

20. If $P(A \cup B) = P(A)$, then $P(A \cap B)$ is equal to

I. 0
II. $P(A)$
III. $P(B)$

A. I only.
B. II only.
C. I and III.
D. II and III.

April/ May 2019 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only